

Column-Analysis

CAMPAIGN_NBR (Type: float64, Unique Values: 0): All missing values. Column should be dropped.

ENGINE_TRACE_NBR (Type: object, Unique Values: 88): 12 missing values. Acts as a reference code; retain for traceability.

TRANSMISSION_TRACE_NBR (Type: object, Unique Values: 88): 12 missing values. Similar use as ENGINE_TRACE_NBR.

ENGINE_SOURCE_PLANT (Type: object, Unique Values: 9): 12 missing values. Should be treated as a categorical variable.

TRANSMISSION_SOURCE_PLANT (Type: float64, Unique Values: 6): 12 missing values. Consider converting to string for consistency.

OPTF_FAMILY_CERTIFICATION (Type: object, Unique Values: 3): 10 missing values. Categorical; impute with mode.

OPTF_FAMILY_EMISSION_SYSTEM (Type: object, Unique Values: 8): 8 missing values. Categorical; impute with mode.

TOTALCOST (Type: float64, Unique Values: 94): 6 missing values. Clean and useful for financial analysis.

CAUSAL_PART_NM (Type: object, Unique Values: 18): Few missing values. Useful for failure analysis; may require cleaning.

LAST_KNOWN_DELVR_TYPE_CD (Type: float64, Unique Values: 11): Low missingness. Should be treated as a categorical code.

Data Cleaning and Analysis Report

1. Dropping Irrelevant Columns

The column CAMPAIGN_NBR was dropped from the dataset as it contained no values across any rows. Retaining an entirely empty column offers no analytical value and was therefore removed.

2. Handling Missing Categorical Values

To address missing values in categorical columns, we utilized the SimpleImputer with the strategy set to 'most_frequent'. The following columns were imputed using this approach:

- CAUSAL_PART_NM
- PLANT
- STATE
- REPAIR_DLR_POSTAL_CD
- VEH_TEST_GRP
- OPTN_FAMILY_CERTIFICATION
- OPTF_FAMILY_EMISSION_SYSTEM
- ENGINE_SOURCE_PLANT
- ENGINE_TRACE_NBR

- TRANSMISSION_TRACE_NBR
- LINE_SERIES

3. Handling Missing Numerical Values

For missing values in numerical columns, the median was chosen due to its robustness against outliers. The same SimpleImputer class (with strategy='median') was used on the following columns:

- TOTALCOST
- TRANSMISSION_SOURCE_PLANT
- LAST_KNOWN_DELVRY_TYPE_CD

4. Text Correction: Non-English Characters

One row in the CORRECTION_VERBATIM column contained Chinese characters. A script was developed to detect such instances and replace them with standardized English correction notes, ensuring consistent language throughout the dataset.

5. Cleaning the CUSTOMER_VERBATIM Column

The CUSTOMER_VERBATIM column contained messy and inconsistent free-text entries. We wrote a preprocessing script to normalize this data, making it suitable for downstream analysis and modeling.

6. Standardizing Text Data

This step addressed three key issues:

- Capitalization: All categorical column values were converted to Title Case.
- Inconsistencies: Erroneous or placeholder values (e.g., '*') were identified and replaced with the most frequent valid entry using SimpleImputer.
- Typos: Column names and values with spelling errors were corrected. Columns affected included:
 - CORRECTION_VERBATIM
 - PLATFORM
 - CAUSAL_PART_NAME
 - PLANT
 - BUILD_COUNTRY
 - STATE
 - COUNTRY_SALE_ISO

7. Outlier Detection and Treatment

Outliers were identified in the following numerical columns using the IQR method:

- REPAIR_AGE
- KILOMETERS
- REPORTING_COST
- TOTALCOST
- LABORCOST

Instead of removing outliers, we applied Winsorization to cap extreme values at the IQR thresholds, preserving data size while reducing distortion.

Visualization Insights and Critical Columns

Top 5 Critical Columns for Stakeholders

1. TOTALCOST

- Why: Central financial metric
- Insight: Helps stakeholders monitor cost efficiency and manage budgets.

2. REPAIR_AGE

- Why: Indicates age at failure
- Insight: Used to assess product lifespan and reliability.

3. LABOR_COST

- Why: Reflects maintenance effort
- Insight: Helps identify high-effort or inefficient repair processes.

4. CAUSAL_PART_NAME

- Why: Shows components responsible for failure
- Insight: Supports quality assurance and recall decisions.

5. KILOMETERS

- Why: Measures usage before failure
- Insight: Useful for studying wear patterns and maintenance scheduling.

These columns were chosen as they cover:

- Cost Drivers (TOTAL_COST, LABORCOST)
- Reliability Indicators (REPAIR_AGE, KILOMETERS)
- Root Cause Analysis (CAUSAL_PART_NAME)

Visualization Insights

1. Total Cost Distribution – A bar plot revealed a roughly normal distribution, suggesting most repair costs cluster around a central value.
2. Labor Cost vs Total Cost – Scatter plot analysis indicated a moderately strong, positive linear relationship, confirming that labor cost significantly contributes to the total cost.
3. Kilometers vs Total Cost – A weakly negative linear relationship was observed, implying that vehicles with higher costs may tend to fail after longer usage periods.

Generated Tags & Key takeaways

Objective:

To generate structured tags from unstructured fields such as correction notes, customer complaints, and part names. This enables easy summarization of failure conditions, components, and actions.

Tag Categories & Examples:

- Components: steering wheel, assembly, cover, horn switch, module
- Symptoms: coming apart, not working, intermittent, inoperative
- Actions: replacement, replaced, checked, repair, diagnostics

Key Highlights:

- Steering wheel assembly is the most frequently tagged component.
- Most common failures relate to physical or electrical malfunction.
- Replacement is the dominant fix strategy across issues.
- The term 'jet black' appears often, suggesting trim-specific issues.